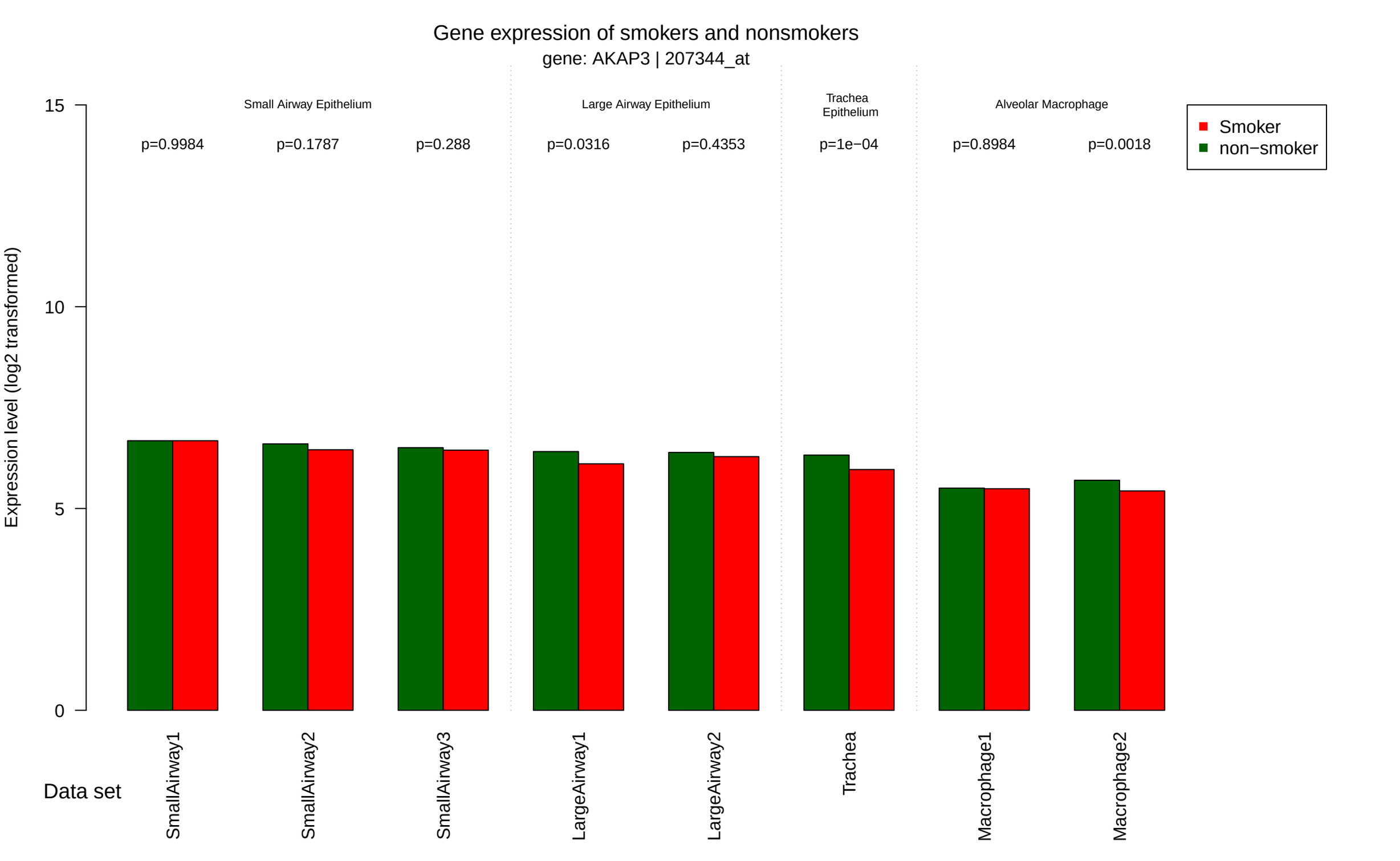
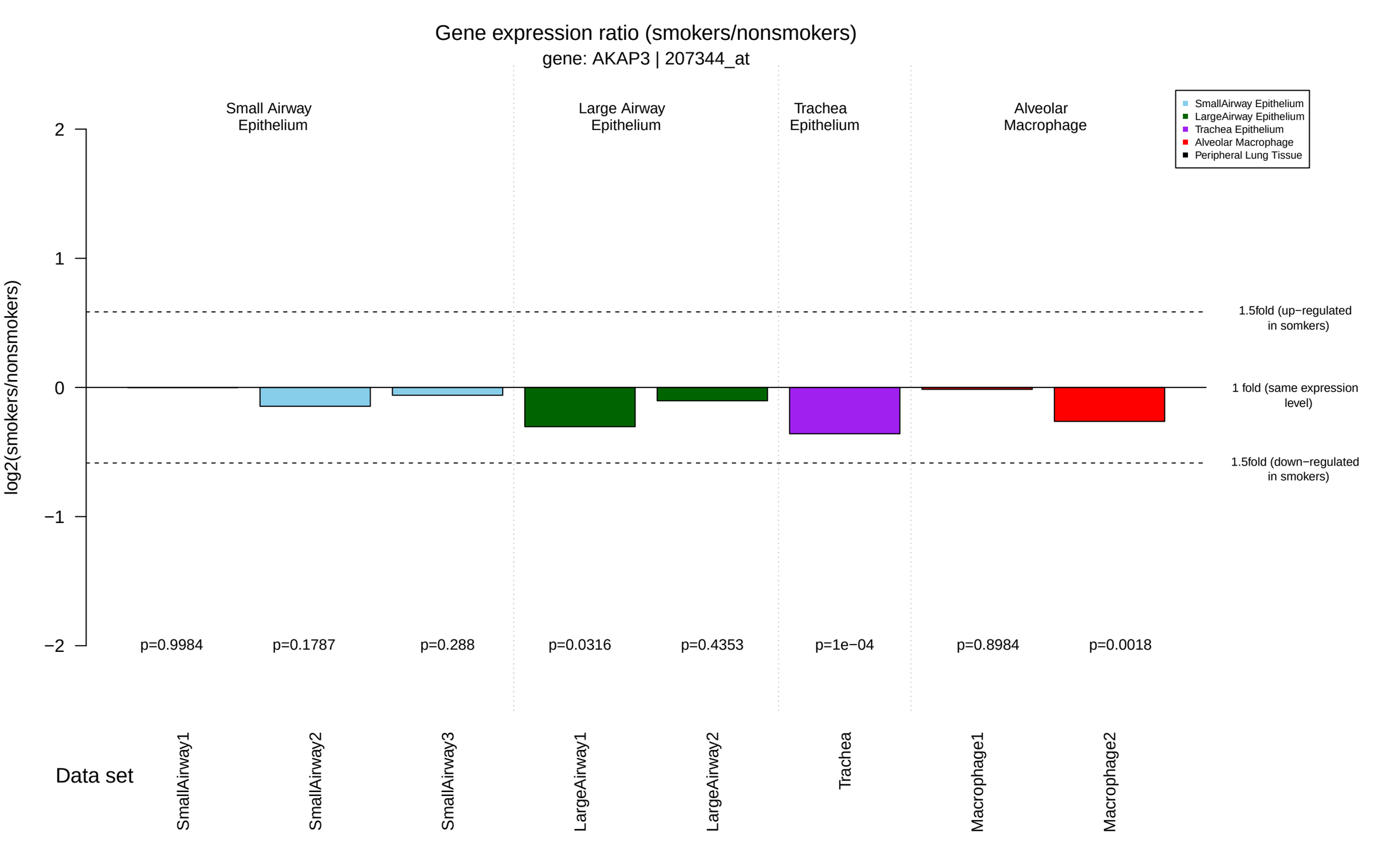
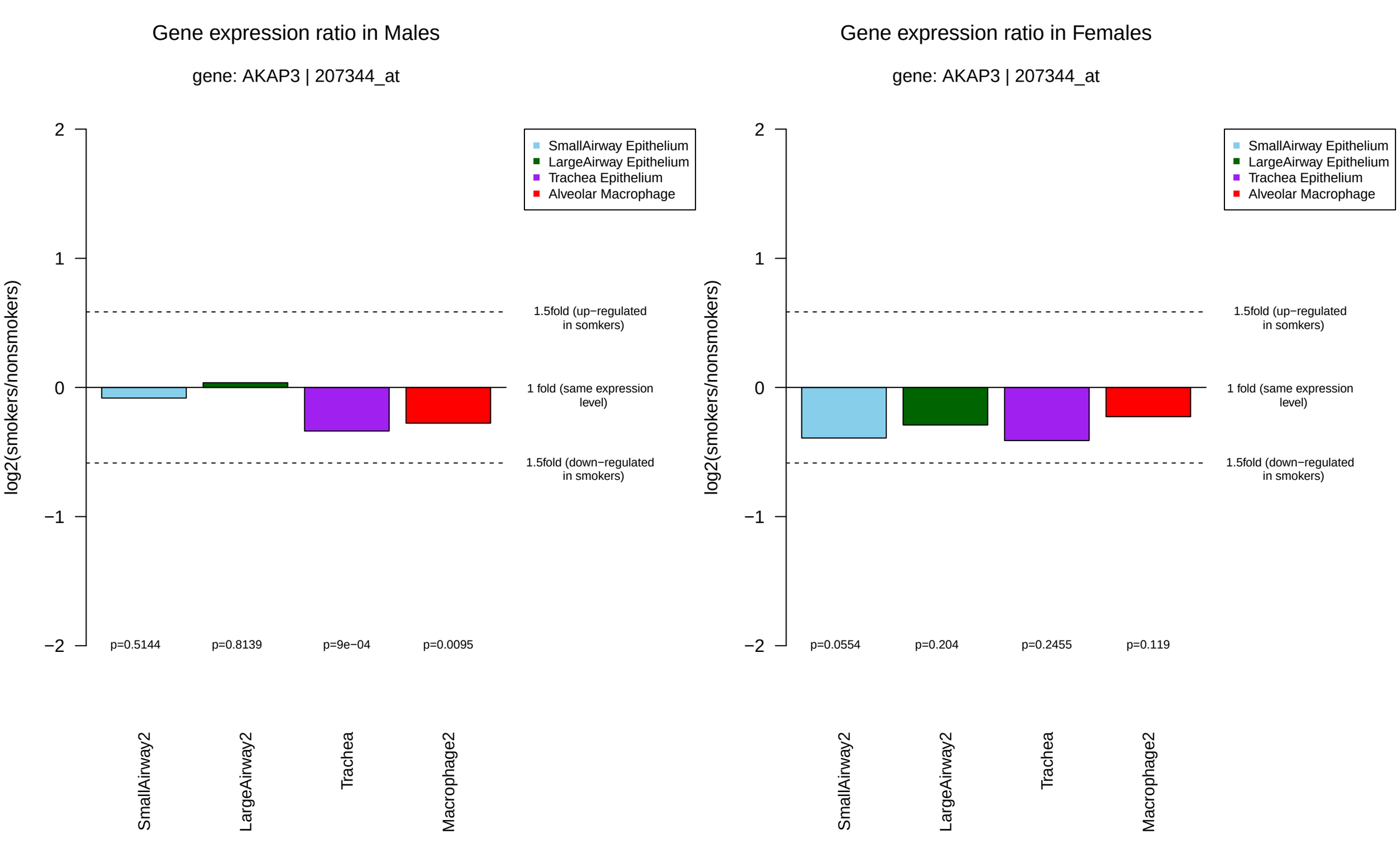
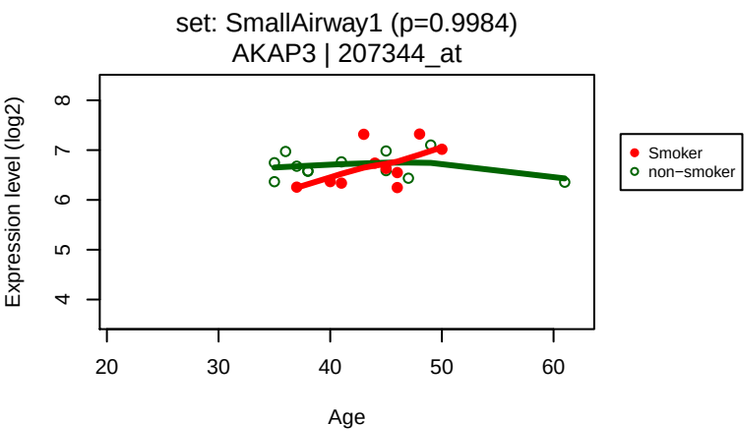
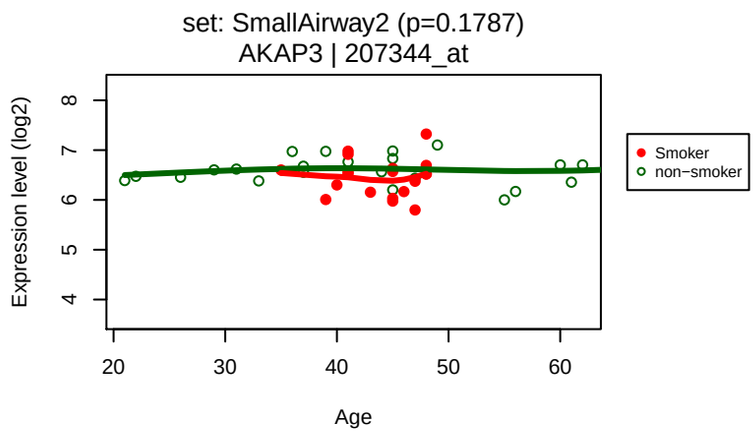
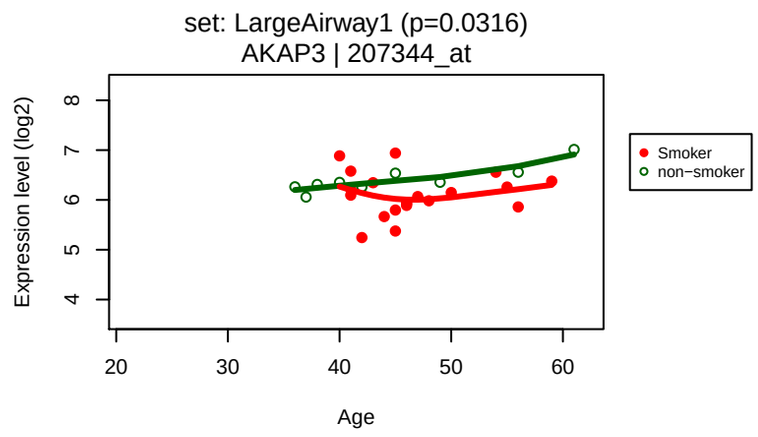

Gene expression of smokers and nonsmokers
gene: AKAP3 | 207344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9984
p=0.1787
p=0.288
p=0.0316
p=0.4353
p=1e−04
p=0.8984
p=0.0018
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

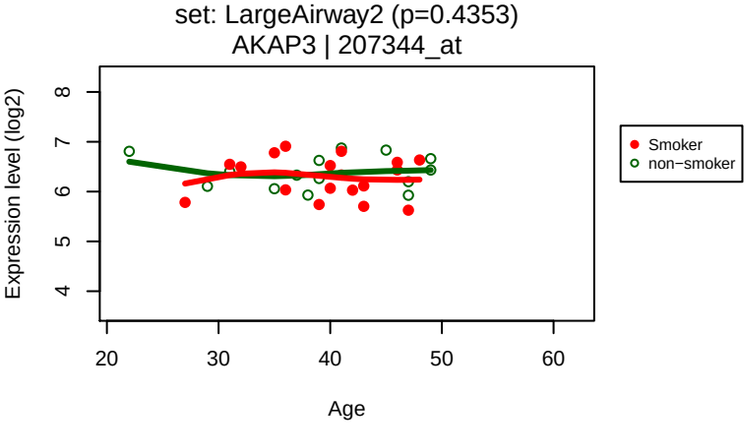

Gene expression ratio (smokers/nonsmokers)
gene: AKAP3 | 207344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9984
p=0.1787
p=0.288
p=0.0316
p=0.4353
p=1e−04
p=0.8984
p=0.0018
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

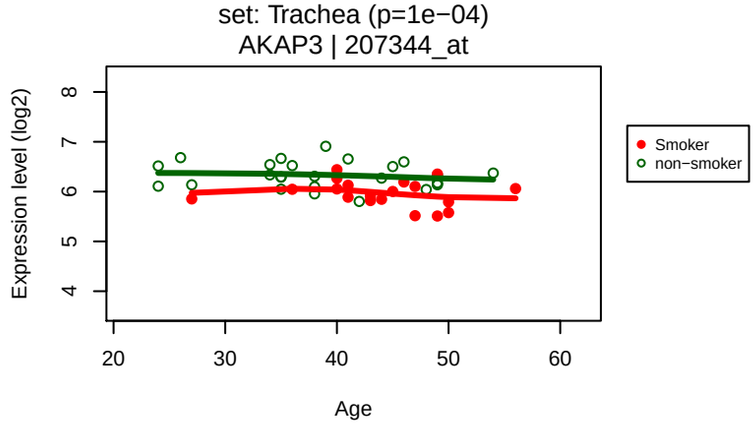

Gene expression ratio in Males
gene: AKAP3 | 207344_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5144
p=0.8139
p=9e−04
p=0.0095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP3 | 207344_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0554
p=0.204
p=0.2455
p=0.119
SmallAirway2
LargeAirway2
Trachea
Macrophage2

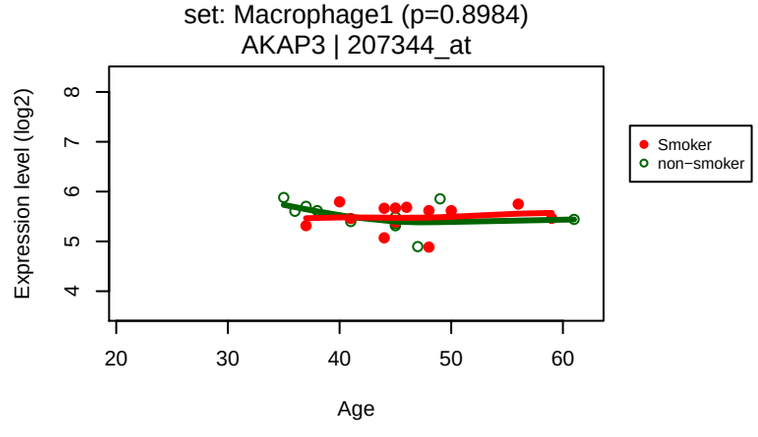

set: SmallAirway1 (p=0.9984)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker

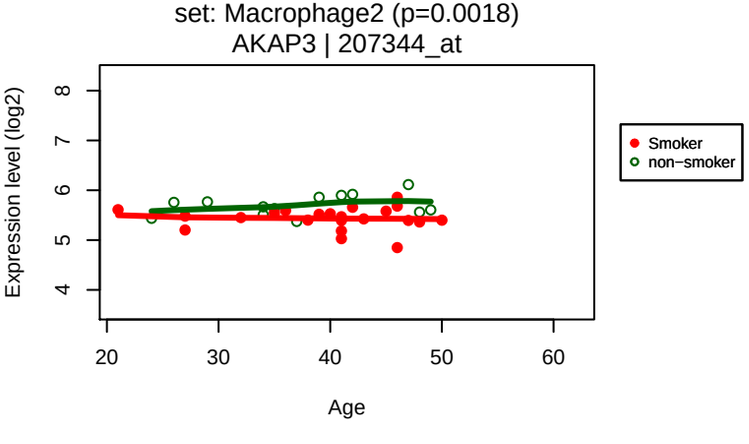

set: SmallAirway2 (p=0.1787)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker

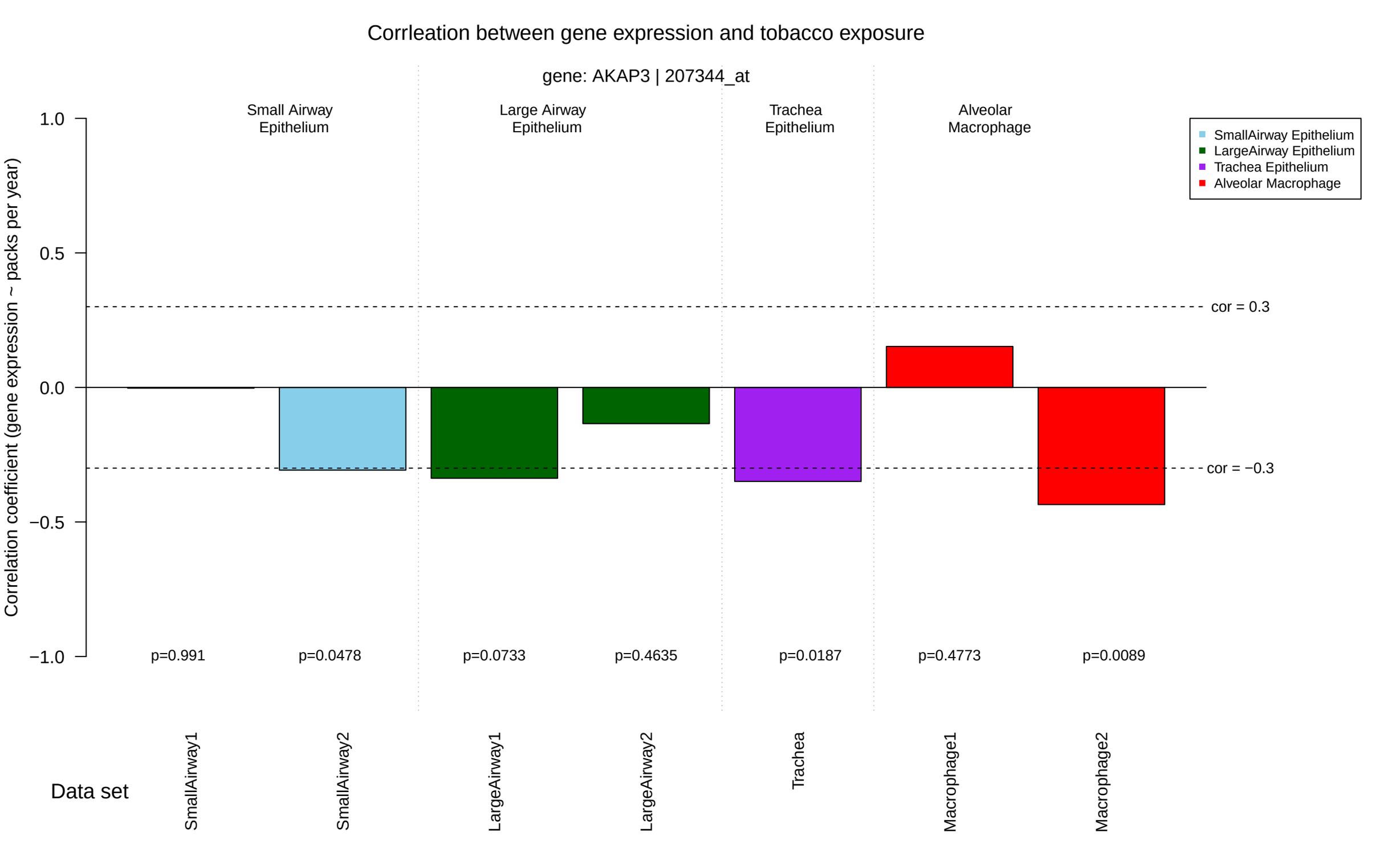

set: LargeAirway1 (p=0.0316)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4353)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8984)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0018)
AKAP3 | 207344_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP3 | 207344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.991
p=0.0478
p=0.0733
p=0.4635
p=0.0187
p=0.4773
p=0.0089
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2